

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

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CO4 -AT2 - Coding Golf Contest

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QUESTIONS**Q1**
Quick Sort
25 marks**Q2**
Bubbled Sorting
25 marks**Q3**
Insertion Sort
25 marks**Q4**
Merger Sort
25 marks

3/4 answered

Q1 Quick Sort 25 marks

Write a program to sort the array 50, 23, 9, 18, 61, 32 using Quick Sort.

Your Code

```
1 #include<stdio.h>
2 void quicksort(int a[],int low,int high)
3 {
4     int i=low,j=high;
5     int pivot=a[(low+high)/2];
6     int temp;
7
8     while(i<=j)
9     {
10         while(a[i]<pivot)
11             i++;
12         while(a[j]>pivot)
13             j--;
14
15         if(i<=j)
16         {
17             temp=a[i];
18             a[i]=a[j];
19             a[j]=temp;
```

Input

stdin input

Output

sorted array :9 18 23 32 50 61

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Bubbled Sorting

Write a program that implements the bubble sort.

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,j,t,a[100];
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(i=0;i<n-1;i++)
9         for(j=0;j<n-1-i;j++)
10             if(a[j]>a[j+1])
11             {
12                 t=a[j];
13                 a[j]=a[j+1];
14                 a[j+1]=t;
15             }
16     for(i=0;i<n;i++)
17         printf("%d ",a[i]);
18     return 0;
19 }
20
21
```

Input5
8 5 3 9 2**Output**

2 3 5 8 9



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Quick Sort
25 marks

Q2
Bubbled Sorting
25 marks

Q3
Insertion Sort
25 marks

Q4
Merger Sort
25 marks

2/4 answered

Write a program sort the elements 9, 5, 1, 4, 3 using Insertion Sort.

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,j,t,a[100],m;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(i=1;i<n;i++)
9     {
10        j=i;
11        while(j>0 && a[j]<a[j-1])
12        {
13            t=a[j];
14            a[j]=a[j-1];
15            a[j-1]=t;
16            j--;
17        }
18    }
19    for(i=0;i<n;i++)
20        printf("%d ",a[i]);
21    return 0;
22 }
```

Input

5
9 5 1 4 3

Output

1 3 4 5 9



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Quick Sort
25 marks

Q2
Bubbled Sorting
25 marks

Q3
Insertion Sort
25 marks

Q4
Merger Sort
25 marks

4/4 answered

Apply Merge Sort to the array 38, 27, 43, 3, 9, 82, 10.

Your Code

```
1 #include<stdio.h>
2 void merge(int a[],int l,int m,int r)
3 {
4     int i=l,j=m+1,k=0,b[20];
5
6     while(i<=m && j<=r)
7         b[k++]=a[i]<a[j]?a[i++]:a[j++];
8     while(i<=m)
9         b[k++]=a[i++];
10    while(j<=r)
11        b[k++]=a[j++];
12    for(i=l,k=0;i<=r;i++)
13        a[i]=b[k++];
14 }
15 void sort(int a[],int l,int r)
16 {
17     if(l<r)
18     {
19         int m=(l+r)/2;
20         sort(a,l,m);
21         sort(a,m+1,r);
22         merge(a,l,m,r);
23     }
24 }
```

Input

stdin input

Output

38 38 10 6553710 -33554178 6881367
3342391

